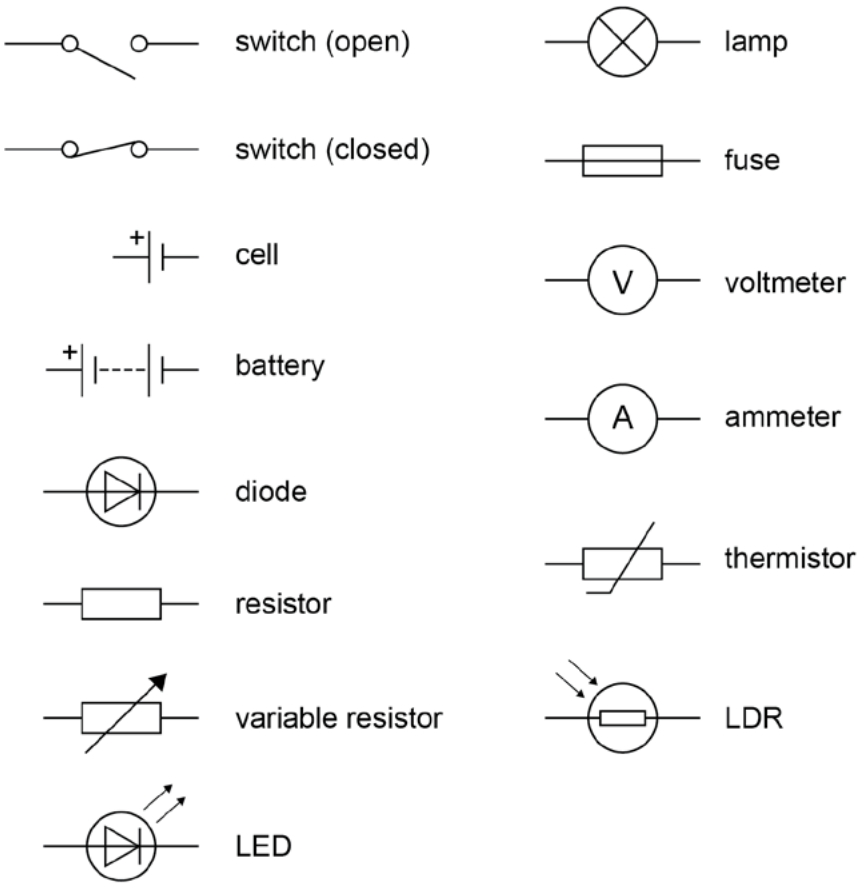
4.2 Electricity

Electric charge is a fundamental property of matter everywhere. Understanding the difference in the microstructure of conductors, semiconductors and insulators makes it possible to design components and build electric circuits. Many circuits are powered with mains electricity, but portable electrical devices must use batteries of some kind.

Electrical power fills the modern world with artificial light and sound, information and entertainment, remote sensing and control. The fundamentals of electromagnetism were worked out by scientists of the 19th century. However, power stations, like all machines, have a limited lifetime. If we all continue to demand more electricity this means building new power stations in every generation – but what mix of power stations can promise a sustainable future?

4.2.1 Current, potential difference and resistance

4.2.1.1 Standard circuit diagram symbols

Content	Key opportunities for skills development
Circuit diagrams use standard symbols.  Students should be able to draw and interpret circuit diagrams.	WS 1.2

4.2.1.2 Electrical charge and current

Content	Key opportunities for skills development
For electrical charge to flow through a closed circuit the circuit must include a source of potential difference. Electric current is a flow of electrical charge. The size of the electric current is the rate of flow of electrical charge. Charge flow, current and time are linked by the equation: charge flow = current × time [$Q = I\,t$] charge flow, *Q*, in coulombs, C current, *I*, in amperes, A (amp is acceptable for ampere) time, *t*, in seconds, s A current has the same value at any point in a single closed loop.	MS 3b, c Students should be able to recall and apply this equation.

4.2.1.3 Current, resistance and potential difference

Content	Key opportunities for skills development
The current (*I*) through a component depends on both the resistance (*R*) of the component and the potential difference (*V*) across the component. The greater the resistance of the component the smaller the current for a given potential difference (pd) across the component. Questions will be set using the term potential difference. Students will gain credit for the correct use of either potential difference or voltage. Current, potential difference or resistance can be calculated using the equation: potential difference = current × resistance [$V = I\,R$] potential difference, *V*, in volts, V current, *I*, in amperes, A (amp is acceptable for ampere) resistance, *R*, in ohms, Ω	MS 3b, c Students should be able to recall and apply this equation.

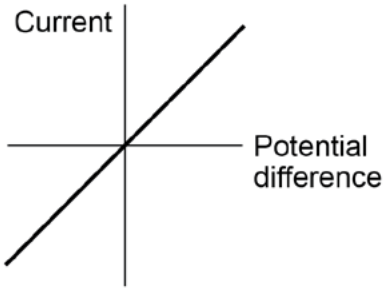
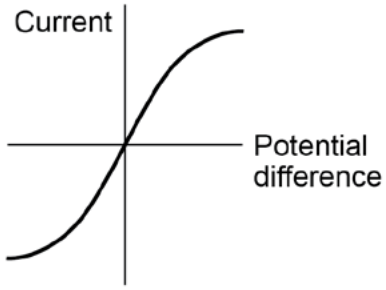
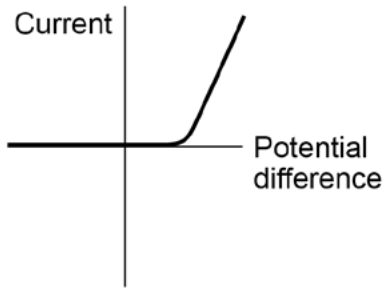
Required practical activity 3: use circuit diagrams to set up and check appropriate circuits to investigate the factors affecting the resistance of electrical circuits. This should include:

- the length of a wire at constant temperature
- combinations of resistors in series and parallel.

AT skills covered by this practical activity: AT 1, 6 and 7.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development.

4.2.1.4 Resistors

Content	Key opportunities for skills development
Students should be able to explain that, for some resistors, the value of *R* remains constant but that in others it can change as the current changes. The current through an ohmic conductor (at a constant temperature) is directly proportional to the potential difference across the resistor. This means that the resistance remains constant as the current changes.  The resistance of components such as lamps, diodes, thermistors and LDRs is not constant; it changes with the current through the component. The resistance of a filament lamp increases as the temperature of the filament increases.  The current through a diode flows in one direction only. The diode has a very high resistance in the reverse direction.  The resistance of a thermistor decreases as the temperature increases. The applications of thermistors in circuits eg a thermostat is required. The resistance of an LDR decreases as light intensity increases.	

Content	Key opportunities for skills development
The application of LDRs in circuits eg switching lights on when it gets dark is required. Students should be able to: • explain the design and use of a circuit to measure the resistance of a component by measuring the current through, and potential difference across, the component • draw an appropriate circuit diagram using correct circuit symbols.	WS 1.2, 1.4 AT 6 Investigate the relationship between the resistance of a thermistor and temperature. Investigate the relationship between the resistance of an LDR and light intensity.
Students should be able to use graphs to explore whether circuit elements are linear or non-linear and relate the curves produced to their function and properties.	WS 1.2, 1.4 MS 4c, d, e

Required practical activity 4: use circuit diagrams to construct appropriate circuits to investigate the I–V characteristics of a variety of circuit elements, including a filament lamp, a diode and a resistor at constant temperature.

AT skills covered by this practical activity: AT 6 and 7.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development.

4.2.2 Series and parallel circuits

Content	Key opportunities for skills development
There are two ways of joining electrical components, in series and in parallel. Some circuits include both series and parallel parts.	
For components connected in series: • there is the same current through each component • the total potential difference of the power supply is shared between the components • the total resistance of two components is the sum of the resistance of each component.	
$R_{total} = R_1 + R_2$	MS 1c, 3b, 3c, 3d
resistance, R, in ohms, Ω	
For components connected in parallel: • the potential difference across each component is the same • the total current through the whole circuit is the sum of the currents through the separate components • the total resistance of two resistors is less than the resistance of the smallest individual resistor.	
Students should be able to: • use circuit diagrams to construct and check series and parallel circuits that include a variety of common circuit components	AT 7
• describe the difference between series and parallel circuits • explain qualitatively why adding resistors in series increases the total resistance whilst adding resistors in parallel decreases the total resistance	
• explain the design and use of dc series circuits for measurement and testing purposes	WS 1.4
• calculate the currents, potential differences and resistances in dc series circuits	MS 1c, 3b, c, d
• solve problems for circuits which include resistors in series using the concept of equivalent resistance.	
Students are **not** required to calculate the total resistance of two resistors joined in parallel.	